

 Library Management System (CS253 Assignment 1)

Course: CS253 - Software Development and Operations

Date: Sunday, March 09, 2025

 Overview

This project implements a Library Management System (LMS) using Object-Oriented Programming (OOP) concepts in C++. The system supports three types of users—Students, Faculty, and Librarians—with specific roles and permissions. It allows users to borrow, return, and manage books efficiently while demonstrating OOP principles like inheritance, polymorphism, encapsulation, and abstraction.

 Features

 User Roles

Role	Borrow Limit	Borrow Duration	Fine	Permissions
Student	3 Books	15 Days	₹10/day overdue	Borrow/Return Books, Pay Fines
Faculty	5 Books	30 Days	No Fine	Borrow/Return Books
Librarian	N/A	N/A	N/A	Manage Books/Users

 Book Management

- Add/Remove books (Librarians only).
- Track book status: "available", "borrowed", "reserved".

 Borrowing Rules

- Borrow limits enforced based on user roles.
- Students must pay overdue fines before borrowing new books.
- Faculty cannot borrow additional books if overdue exceeds 60 days.

 Returning Rules

- Update book status to "available" upon return.
- Calculate fines for students based on overdue days.

 File Persistence

Save and load data (books, accounts) using JSON files (books.json, accounts.json).

Project Structure

CS253_Project/

```
|—— main.cpp          # Main program logic
|—— classes.h         # Class definitions
|—— library.h, library.cpp # Library management logic
|—— users.h, users.cpp  # User role implementations
|—— transaction.h, transaction.cpp # Fine calculation logic
|—— cli/
|   |—— home.h         # Main menu logic
|   |—— login_lib.h    # Librarian login logic
|   |—— login_stud_fac.h # Student/Faculty login logic
|—— data/
|   |—— books.json     # Book data
|   |—— accounts.json  # User account data
|—— json.hpp          # JSON library (from nlohmann/json)
```

Setup Instructions

Step 1: Install Dependencies

Download the JSON for Modern C++ library. Place the json.hpp file in your project directory.

Step 2: Compile the Code

Run the following command in your terminal:

```
g++ -std=c++17 main.cpp library.cpp users.cpp transaction.cpp -o  
library_management
```

Step 3: Run the Program

Execute the compiled program:

```
./library_management
```

Data Files

- ◆ accounts.json

Tracks user accounts.

- ◆ books.json

Tracks book details.

Usage Instructions

Upon running the program, you will see a menu with options:

1. Borrow Book
2. Return Book
3. Pay Fine
4. Add Book (Librarian)
5. Remove Book (Librarian)
6. Exit
7. Increment Simulated Date
8. View Borrowing History
9. Add User (Librarian)
10. Remove User (Librarian)

Enter choice:

Select an option by entering its corresponding number.

Important Notes

- Students must pay fines before borrowing new books.
- Faculty members receive warnings for overdue books exceeding 60 days.
- Librarians can manage books and users but cannot borrow or return books.